In Support of Securing Fen Protections via Regulation 87 to Combat Climate Change

To: Colorado's Water Quality Control Commission

The protection of Colorado's fen ecosystems is a vital strategy in the fight against climate change. Fens store large amounts of carbon in their soils, and the slightest disturbances in these ecosystems can cause a significant and ongoing release of harmful greenhouse gases into the atmosphere. Infrastructure development, mining, and water diversion projects have all posed serious threats to these fragile habitats in recent years. As Colorado's Water Quality Control Commission drafts Regulation 87, now is the time to designate fens as protected state resources and fund fen identification and stewardship efforts.

What is a Fen?

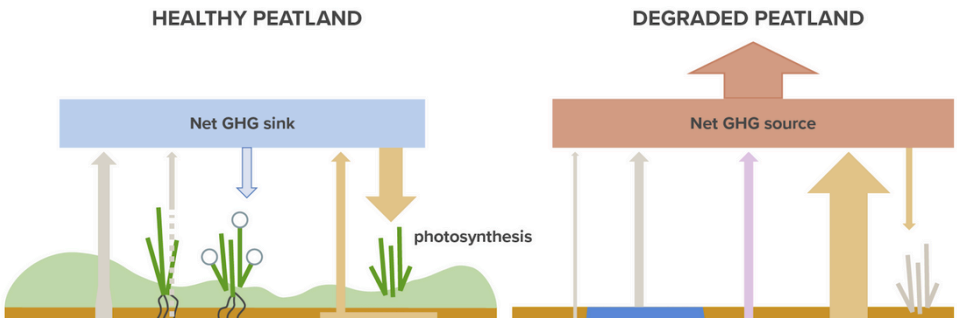
Colorado is home to a unique and rare type of high-elevation peatland, known as fens. Fens form by accumulating peat at a very slow rate of around 20cm per 1,000 years, making them irreplaceable ecosystems. Some of Colorado's fens are over 10,000 years old and are often called "old-growth wetlands" (Helt, 2025). They are home to a variety of sensitive plant and animal species, including the state-listed endangered Boreal Toad (Culver & Smith, 2019). Colorado's fens also function as a secondary water source, absorbing initial snowmelt and gradually returning it to waterways once the snow has disappeared (Helt, 2025). Healthy fen and wetland ecosystems play a crucial role in mitigating the effects of drought as well as wildfire. The water stored in these ecosystems create fire breaks and helps reduce wildfire intensity (Benko, Wurster, & Balentine, 2024).

Hayden Meadow Extreme Rich Fen (Lake County, CO). Photo by Michael Menefee, reproduced from Culver & Smith (2019, p. 39)

In addition to their water storage, fire resiliency, and biodiversity benefits, fens are also carbon sinks, storing large amounts of carbon in their peat-rich soils. While peatlands only cover approximately 4% of the Earth's surface, they contain two times the amount of carbon than the planet's forests (United Nations Environment Programme, 2022). Colorado's wetlands are estimated to represent less than 3% of the state's land area, with fens making up only a small fraction of that total (Colorado Natural Heritage Program, n.d.). With small footprints but large carbon stores, protecting fens is feasible and highly impactful.

What's at Stake?

Despite the critical role fens play as carbon sinks, these ecosystems face continued threat in Colorado from human activities and development. Colorado's fens have already been impacted by road and infrastructure development (Lemly et al., 2018) as well as mining (Kuenzi, n.d.). The proposed Whitney dam project currently threatens a large fen complex in Eagle County (Sakas, 2021).



Source: IUCN UK Peatland Programme. (2024, July 10). *New briefing addresses the peatlands and methane debate.*

Once degraded, fens can take hundreds to thousands of years to recover and the restoration process is uncertain and complex (Helt, 2025). The greenhouse gases (GHGs) that escape from the soil when fens are disrupted can likewise remain in the atmosphere for centuries or longer, increasing global temperatures and amplifying the effects of climate change.

In 2002, the USFS Rocky Mountain Region designated fens as irreplaceable resources and the Colorado Natural Heritage Program emphasizes that "every reasonable effort should be made to avoid impacting this habitat" (Culver & Smith, 2019). Yet proposals that affect fens continue, and many fens occur on land owned beyond USFS jurisdiction, creating protection gaps without clear statewide standards.

Key Findings

- Fens are irreplaceable natural resources that take thousands of years to form
- Fen ecosystems store carbon and water, increase fire resiliency, and promote biodiversity
- Fen restoration is difficult, slow, and uncertain
- Colorado's fens are under threat from human impacts
- If disturbed, fens release harmful greenhouse gases into the atmosphere that persist for centuries
- Fen protection is an exceptionally low-cost climate mitigation strategy that delivers broad benefits

Policy Recommendations

Colorado's HB24-1379 requires a state dredge-and-fill permitting program and rulemaking by Dec 31, 2025. This is a pivotal chance to codify fen-specific state protections in Regulation 87.

Recommendation 1: Categorical Protection for Fens

Adoption of a strict avoidance standard that designates fens and their groundwater contributing area as off-limits to dredge/fill, draining, and general disturbances due to infrastructure development, consistent with USFS guidance. Avoidance of fens should be the default and "fen relocation" as mitigation should be prohibited. Compensatory mitigation cannot apply to fens as there is no practical alternative achievable on a human timescale.

Recommendation 2: Fen Stewardship

Establish and fund management plans for all currently identified fens in the state of Colorado. Prioritize conservation planning in areas where human development is in close proximity with documented confirmed fen and potential fen complexes according to the Colorado Wetland Inventory Mapping Tool. Standardized statewide management strategies ensure consistent and continued protection of fens across jurisdictions.

Recommendation 3: Fen Database Upkeep, Identification, and Mapping

Fund the Colorado Natural Heritage Program to continue to improve and maintain the Colorado Wetland Inventory Mapping Tool, with a priority on identifying and mapping fens. Set minimum mapping standards to include fen surface area and contributing groundwater area. A complete and thorough fen database ensures administrators consult a single, authoritative source before permitting any ground disturbance.

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